# Surface area of solids

SURFACE AREA IS THE SPACE OCCUPIED BY A SHAPE'S OUTER SURFACES.

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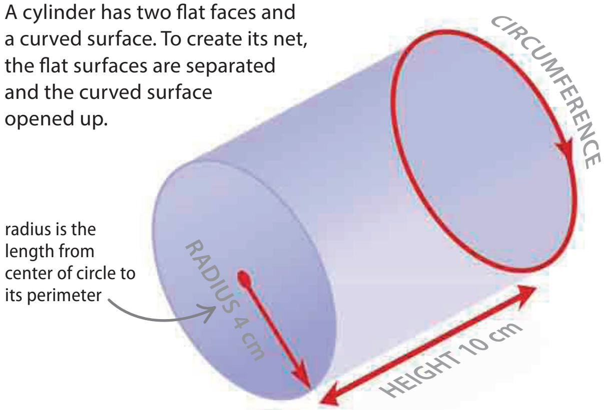
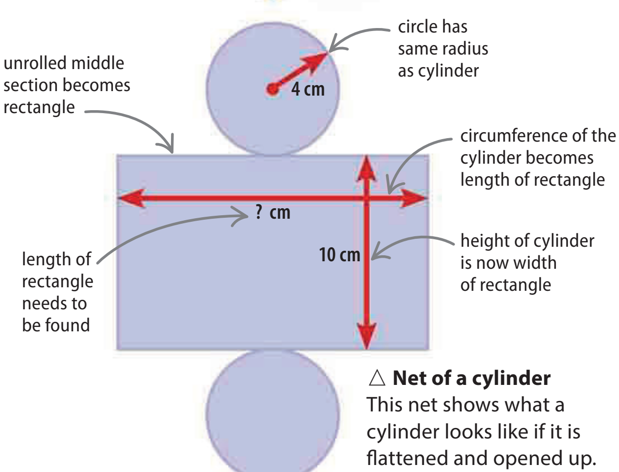
**For most solids, surface area can be found by adding together the areas of its faces. The sphere is the exception, but there is an easy formula to use.**

## Surfaces of shapes

For all solids with straight edges, surface area can be found by adding together the areas of all the solid's faces. One way to do this is to imagine taking apart and flattening out the solid into two-dimensional shapes. It is then straightforward to work out and add together the areas of these shapes. A diagram of a flattened and opened out shape is known as its net.

▷ **Cylinder**
A cylinder has two flat faces and a curved surface. To create its net, the flat surfaces are separated and the curved surface opened up.




△ **Net of a cylinder**
This net shows what a cylinder looks like if it is flattened and opened up. It consists of a rectangle and two circles.

## Finding the surface area of a cylinder

Breaking the cylinder down into its component parts creates a rectangle and two circles. To find the total surface area, work out the area of each of these and add them together.

formula for area of circle

$$\text{Area} = \pi \times r^2$$

area of circle

$$3.14 \times 4 \times 4 = \mathbf{50.24\ cm^2}$$

**The area of the circles** can be worked out using the known radius and the formula for the area of a circle. π (pi) is usually shortened to 3.14, and area is always expressed in square units.

formula for circumference

$$\text{Circumference} = 2 \times \pi \times r$$

circumference of cylinder

$$2 \times 3.14 \times 4 = \mathbf{25.12\ cm}$$

**Before the area of the rectangle can be found**, it is necessary to work out its width—the circumference of the cylinder. This is done using the known radius and the formula for circumference.

length of rectangle = circumference of cylinder

width of rectangle = height of cylinder

area of rectangle

$$25.12 \times 10 = \mathbf{251.2\ cm^2}$$

**The area of the rectangle** can now be found by using the formula for the area of a rectangle (length × width).

surface area of cylinder

$$50.24 + 50.24 + 251.2 = \mathbf{351.68\ cm^2}$$

**The surface area** of the cylinder is found by adding together the areas of the three shapes that make up its net—two circles and a rectangle.